

$(S_L, a_S, 10, S_L) \rightarrow (S_L, a_B, 20, S_H) \rightarrow (S_H, a_S, 0, S_H)$

1.1

$$Q(s, a) = Q(s, a) + \alpha [r + \gamma \min_{a'} Q(s', a') - Q(s, a)]$$

$$\begin{array}{cc} & S & B \\ \begin{array}{c} L \\ H \end{array} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{array}$$

$$Q(s_H, a_S) = Q(s_H, a_S) + \alpha [0 + \gamma \min_{a'} Q(s_H, a') - Q(s_H, a_S)]$$

1.2 Yes the SARSA update would differ

$$Q(s_H, a_S) = Q(s_H, a_S) + \alpha [0 + \gamma Q(s_H, a_B) - Q(s_H, a_S)]$$

Because it uses the Q value of the next state and action actually taken

1.3 Yes, both SARSA and Q -learning are bootstrapping methods because they update the value of a state-action pair using estimates of future returns rather than waiting for the final outcome of an episode.

2.1 $\epsilon = 0.2$ $\begin{cases} 80\% \text{ chooses best action} \\ 20\% \text{ explores randomly} \end{cases}$

$$Q(s_H, a_S) = 5$$

$$Q(s_H, a_B) = 15$$

Greedy action $\rightarrow$ the one with lowest cost $\rightarrow a_S$ (since $5 < 15$)

$$P(a_S) = 0.8 + 0.2 \times 1/2 = 0.9$$

$$P(a_B) = 0.2 \times 1/2 = 0.1 = 10\%$$

2.2 The risk of decreasing ϵ to 0 too quickly is that the agent will stop exploring actions and rely only on its current estimates of cost. This can cause it to settle on suboptimal policy, because it may miss actions or sequence of actions that lead to lower cumulative cost in the long run. Early exploration is essential to accurately learn the true value of actions and avoid shortsighted decisions

3. Based on the trajectory, the only (S_L, a_S) transition goes to S_L once and never to S_H

Transition Count Table:

Current state	Action	S_L	S_H
S_L	a_S	1	0

$$P(s' | s, a) = \frac{\text{Count}(s, a \rightarrow s')}{\sum_{s''} \text{Count}(s, a \rightarrow s'')}$$

$$P(S_L | S_L, a_S) = \frac{1}{1} = 1$$

$$P(S_H | S_L, a_S) = \frac{0}{1} = 0$$

4.1

$$S_L \min(40, 30) \rightarrow a_B$$

$$S_H \min(10, 5) \rightarrow a_B$$

New deterministic policy

$$\pi(S_L) = a_B \quad \pi(S_H) = a_B$$

4.2

$$\gamma = 1$$

$$P(a_S | S_L) = \frac{e^{10 \cdot 40}}{e^{10 \cdot 40} + e^{10 \cdot 30}} = \frac{e^{100}}{e^{100} + 1}$$

